SAMVID OS DOCUMENTATION

SYSTEM WORKFLOW VISUAL

Generated from markdown with browser rendering so workflow diagrams, structure, and formatting are preserved for sharing and presentation.

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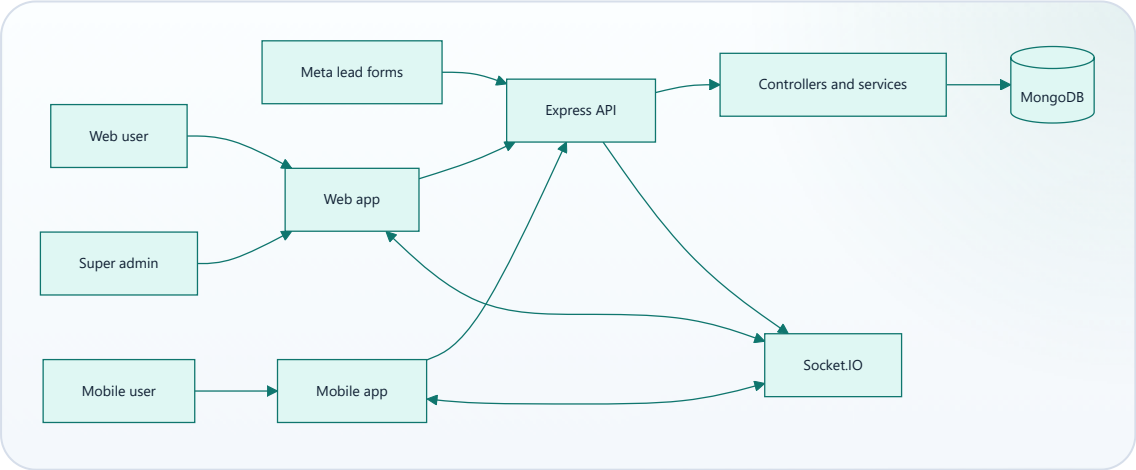
This document gives a visual view of how the major Samvid OS functions work across web, mobile, backend, realtime events, and tenant operations.

It is intentionally business-flow focused. For file-by-file technical coverage, use:

- Web Frontend Pages and Components
- Mobile Screens and Components
- Backend Functionality and API Surface

1. System Map



Core idea

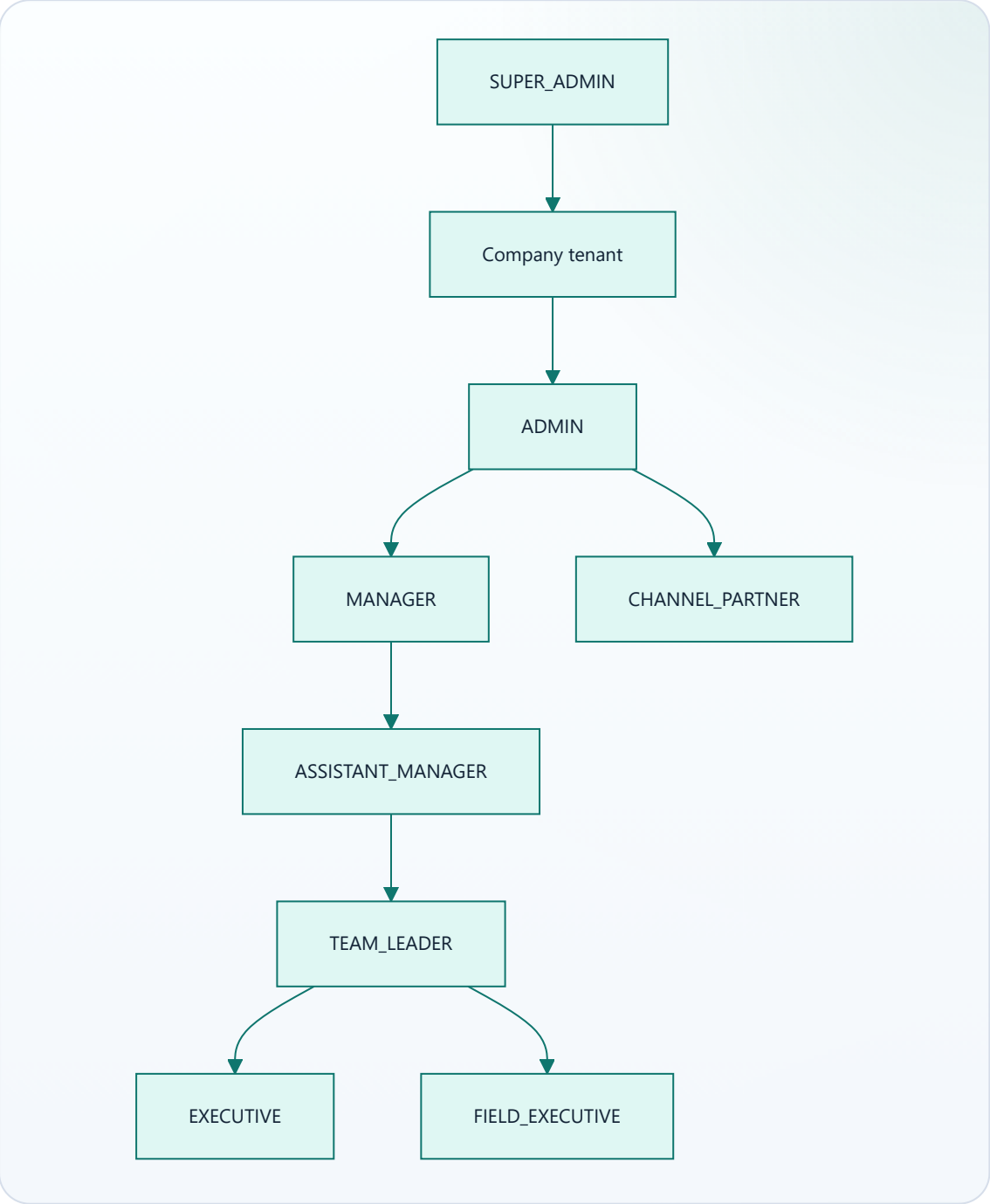
- Web and mobile both use the same backend domain logic.
- API requests handle auth, role gating, tenant scope, and persistence.
- Socket.IO handles realtime chat, call, alert, and approval signals.
- MongoDB stores tenants, users, leads, inventory, chat, targets, and subscriptions.

Primary source anchors

- `frontend/src/App.jsx`
- `mobile/src/navigation/RootNavigator.tsx`

- `backend/src/app.js`
- `backend/src/server.js`

2. Role and Tenant Hierarchy



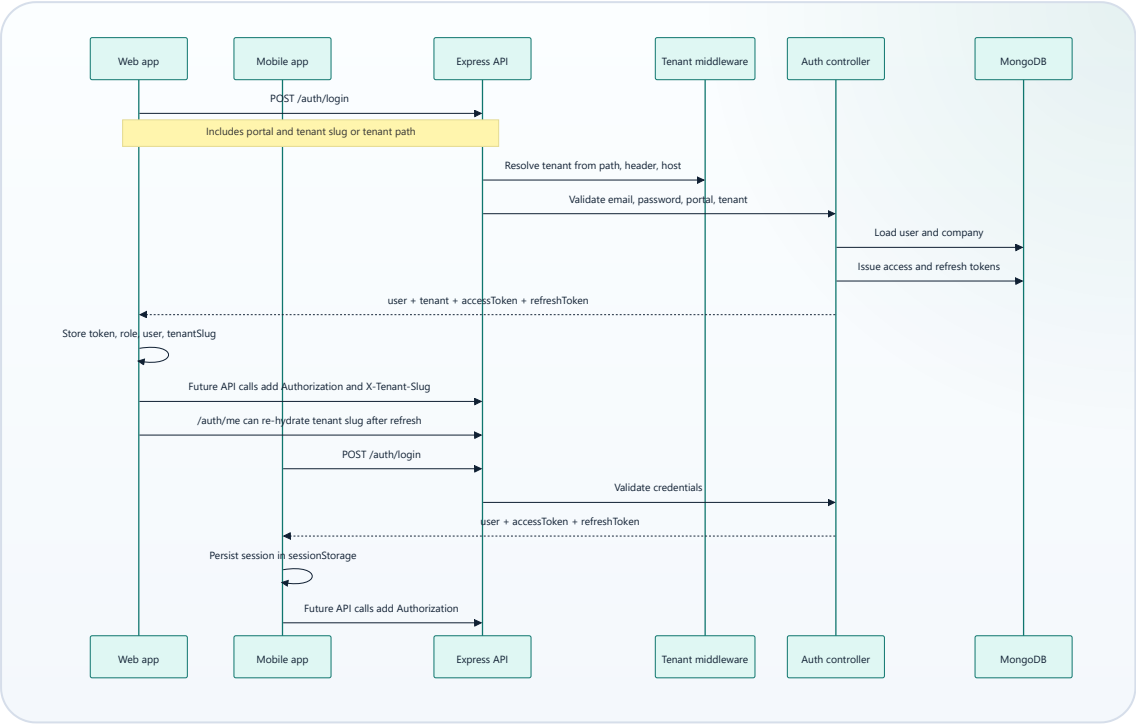
What this means operationally

- `SUPER_ADMIN` works at platform level across all tenants.
- `ADMIN` owns one company tenant and controls approvals, users, and tenant settings.
- Management roles control teams and lead distribution inside the tenant hierarchy.
- Executives and field executives operate on assigned work.
- Channel partners remain tenant-scoped but with tighter access.

Primary source anchors

- `backend/src/constants/role.constants.js`
- `backend/src/routes/user.routes.js`
- `backend/src/routes/saas.routes.js`

3. Login, Tenant Resolution, and Session Flow



Web-specific runtime rules

- The web app keeps tenant-aware URLs like `/<tenant-slug>/leads`.
- Session restore can redirect the browser back into the tenant-prefixed route space.
- Inactivity timeout and theme settings are applied on the client shell.

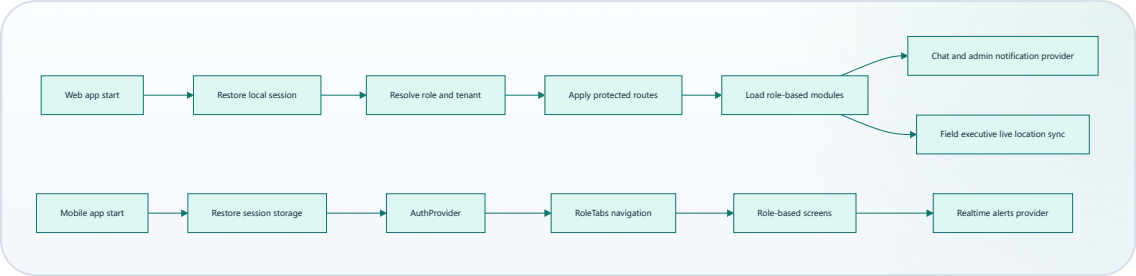
Mobile-specific runtime rules

- Mobile restores the stored session on app start.
- If session expiry is reached in background or refresh fails, the app clears session and returns to login.

Primary source anchors

- `backend/src/middleware/tenant.middleware.js`
- `backend/src/controllers/auth.controller.js`
- `frontend/src/components/auth/Login.jsx`
- `frontend/src/utils/tenantRouting.js`
- `frontend/src/services/api.js`
- `mobile/src/context/AuthContext.tsx`
- `mobile/src/services/api.ts`

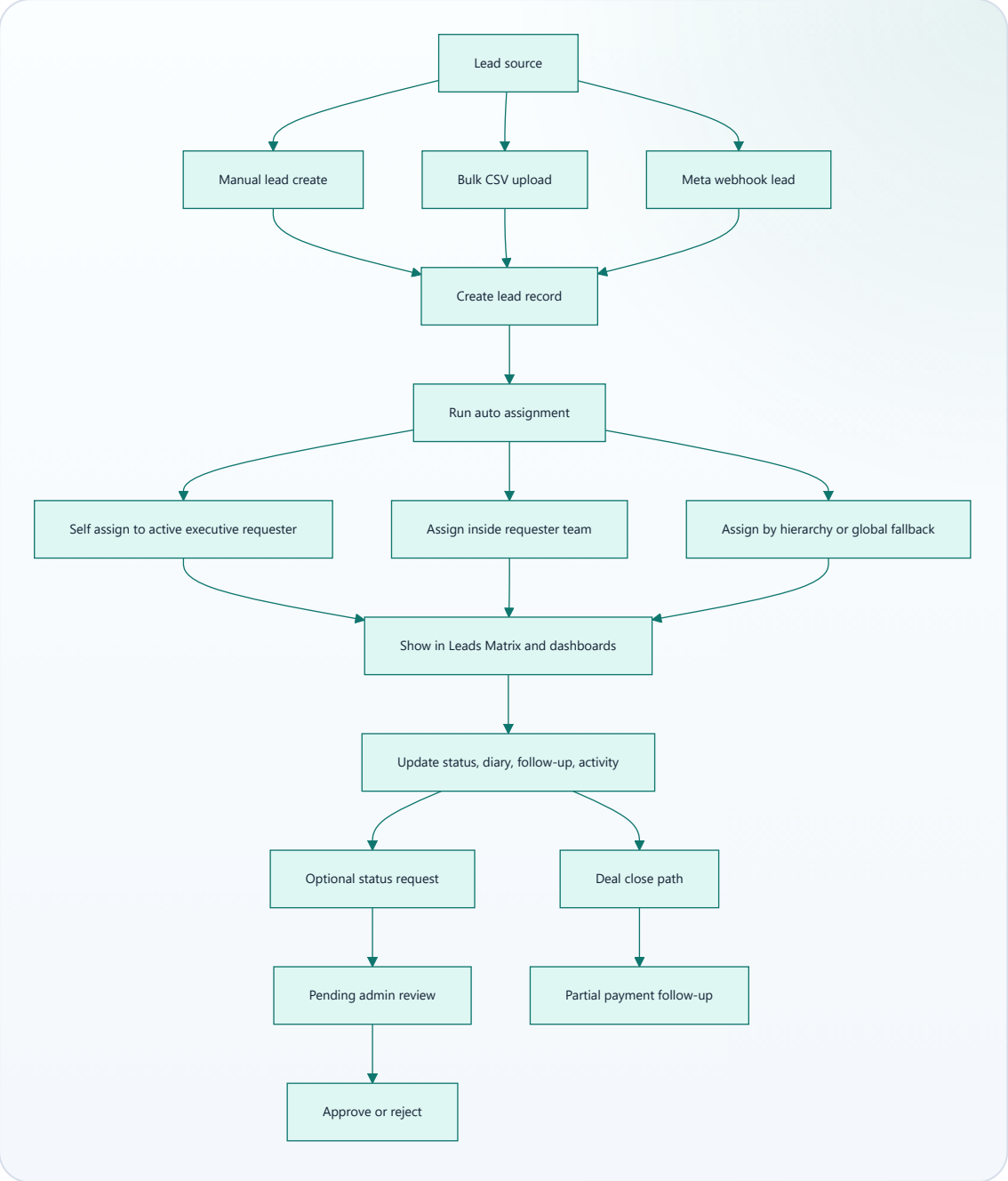
4. Client Runtime Flow



Primary source anchors

- `frontend/src/App.jsx`
- `frontend/src/context/chatNotificationProvider.jsx`
- `mobile/src/navigation/RootNavigator.tsx`
- `mobile/src/navigation/RoleTabs.tsx`
- `mobile/src/context/RealtimeAlertsContext.tsx`

5. Lead Lifecycle



Flow explanation

- Leads enter the system from manual creation, admin bulk upload, or Meta webhook ingestion.
- New leads are auto-assigned using hierarchy-aware balancing logic.
- Users then work the lead through lead details, follow-ups, diaries, and activity tracking.
- Sensitive state transitions can enter a status-request approval path.
- Close and payment flows can trigger admin notifications and additional follow-up work for remaining collection.

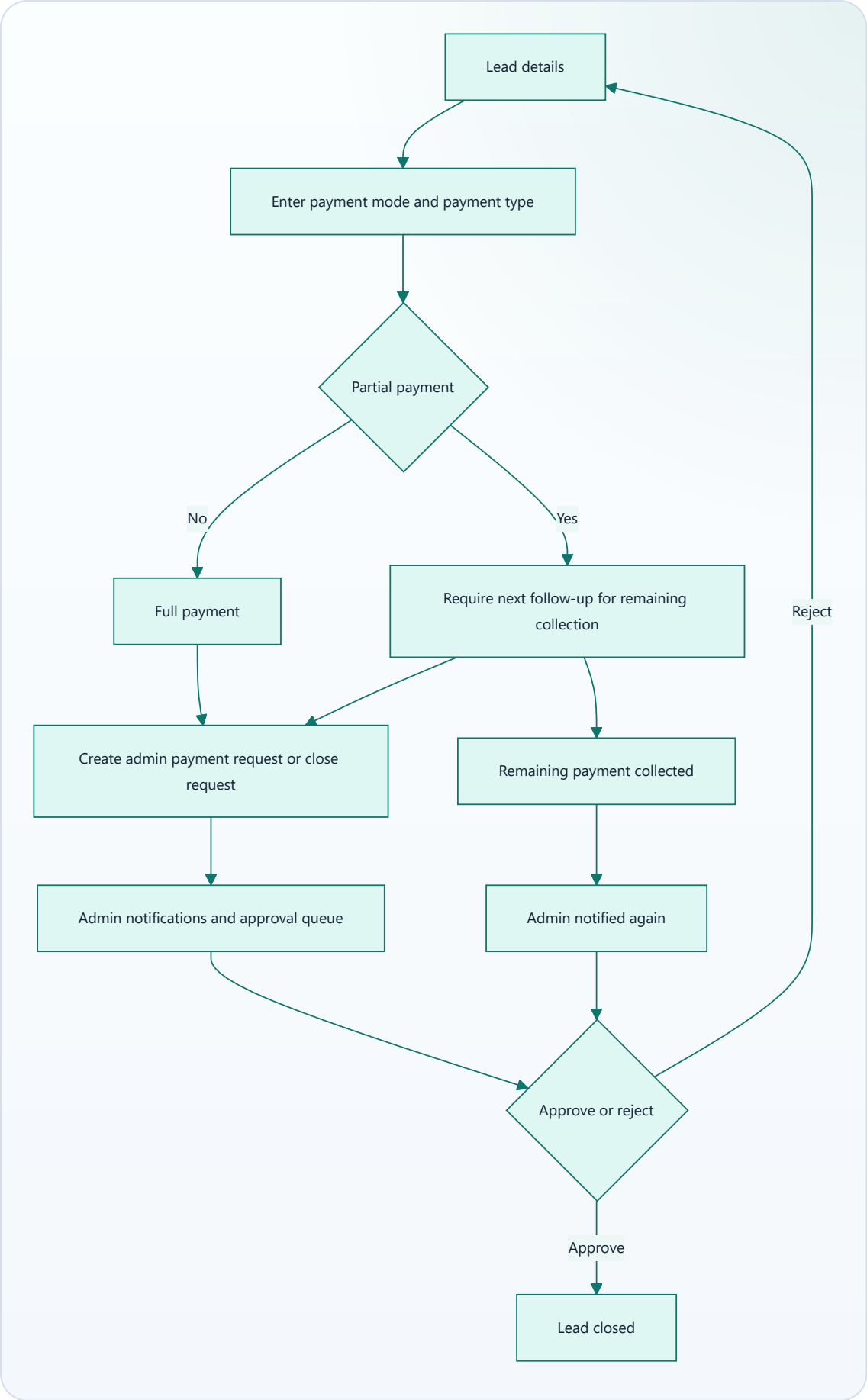
Assignment logic

- If an active executive or field executive creates the lead, the lead can self-assign.
- If a management user creates the lead, the system tries to assign within that reporting tree.
- If no local team candidate is best, the service falls back to hierarchy/global balancing.

Primary source anchors

- `backend/src/routes/lead.routes.js`
- `backend/src/controllers/lead.controller.js`
- `backend/src/services/leadAssignment.service.js`
- `frontend/src/modules/leads/LeadsMatrix.jsx`
- `mobile/src/modules/leads/LeadsMatrixScreen.tsx`
- `mobile/src/modules/leads/LeadDetailsScreen.tsx`

6. Deal Closure and Payment Approval Flow



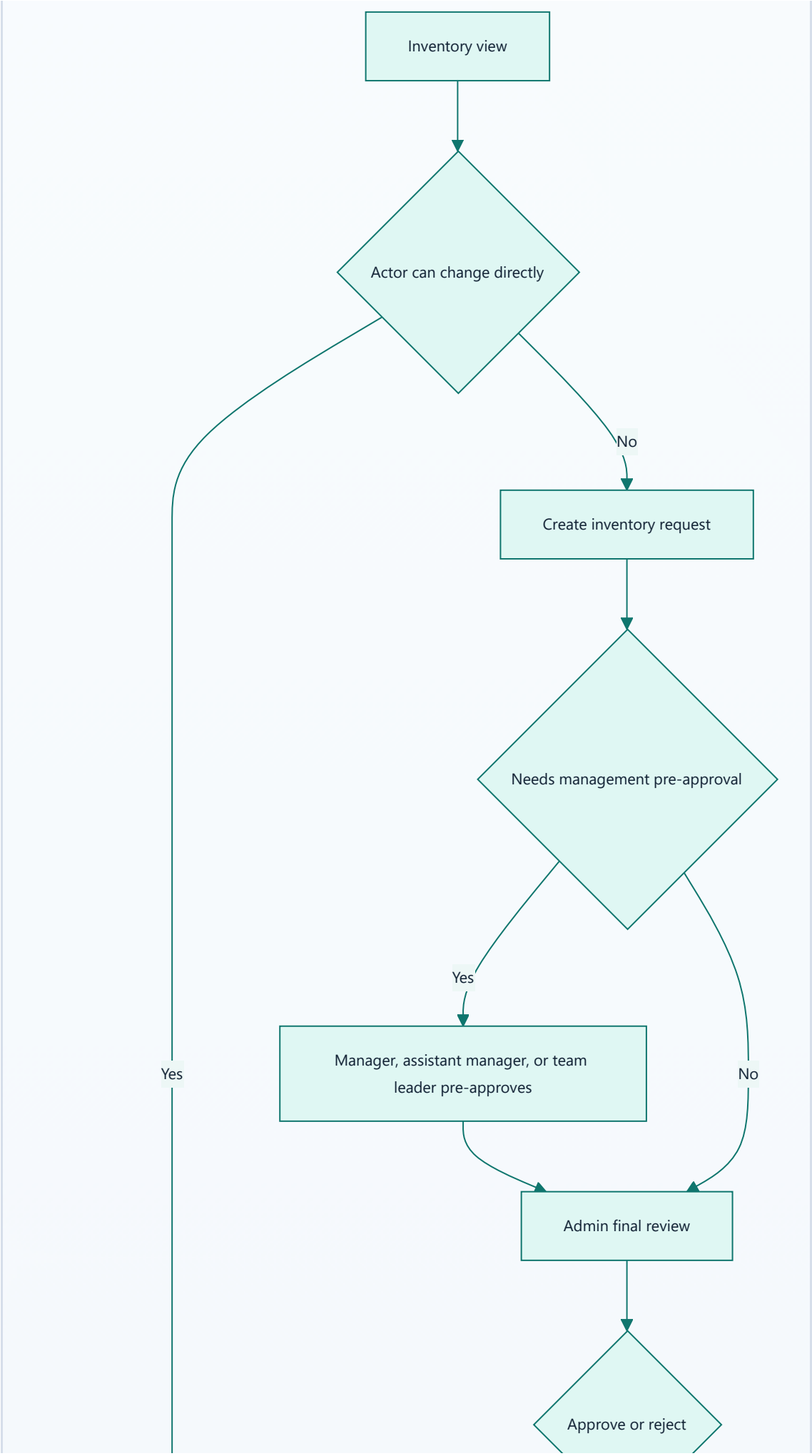
Key business rules

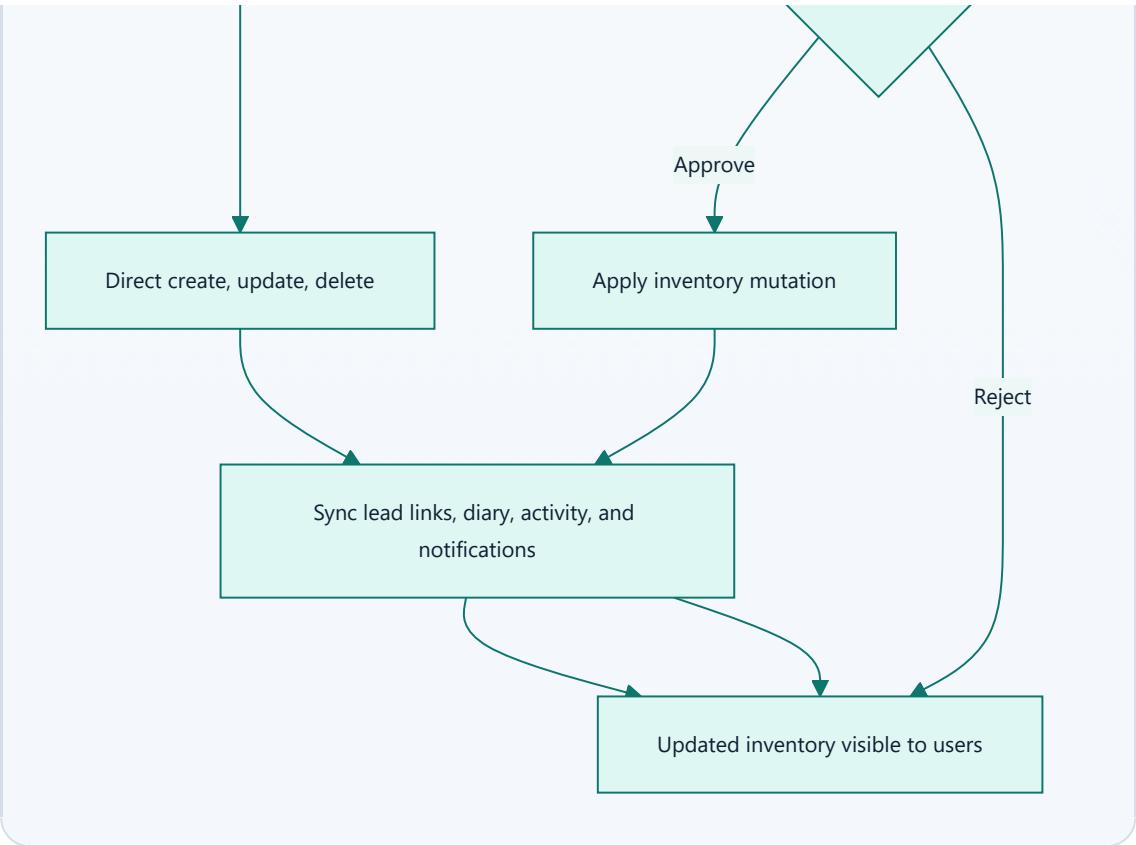
- Non-cash close paths require a payment reference.
- Partial payments require a positive remaining amount.
- Partial-payment close flows must keep the collection loop open through a next follow-up.
- Admin receives realtime events for payment approval, deal closure, and remaining-payment collection.

Primary source anchors

- `backend/src/controllers/lead.controller.js`
- `frontend/src/modules/admin/AdminNotifications.jsx`
- `frontend/src/context/chatNotificationProvider.jsx`
- `mobile/src/modules/notifications/NotificationsScreen.tsx`
- `mobile/src/context/RealtimeAlertsContext.tsx`

7. Inventory Workflow


Inventory view
Actor can change directly
Yes
No
Create inventory request
Needs management pre-approval
Yes
No
Manager, assistant manager, or team leader pre-approves
Admin final review
Approve or reject



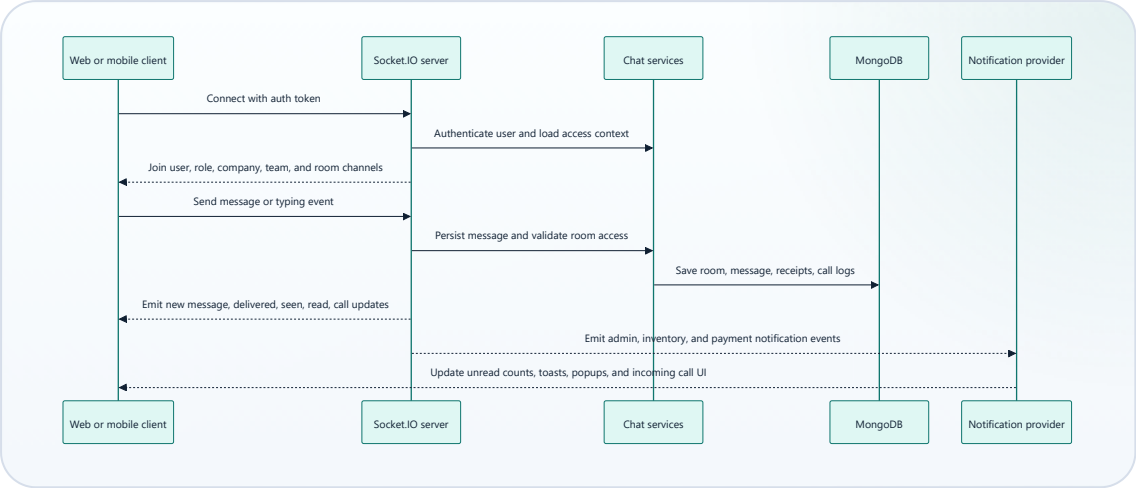
Important status-specific rules

- `Blocked` requests require a lead selection and a diary note.
- `Sold` requests require `saleDetails`, including payment information and the linked lead.
- Review events emit realtime notifications to admin, role, company, and team rooms.

Primary source anchors

- `backend/src/routes/inventory.routes.js`
- `backend/src/routes/inventoryRequest.routes.js`
- `backend/src/controllers/inventory.controller.js`
- `backend/src/controllers/inventoryRequest.controller.js`
- `backend/src/controllers/inventoryApproval.controller.js`
- `backend/src/services/inventoryWorkflow.service.js`
- `backend/src/services/inventoryNotification.service.js`
- `frontend/src/modules/inventory/AssetVault.jsx`
- `mobile/src/modules/inventory/AssetVaultScreen.tsx`

8. Realtime Chat, Alerts, and Calling



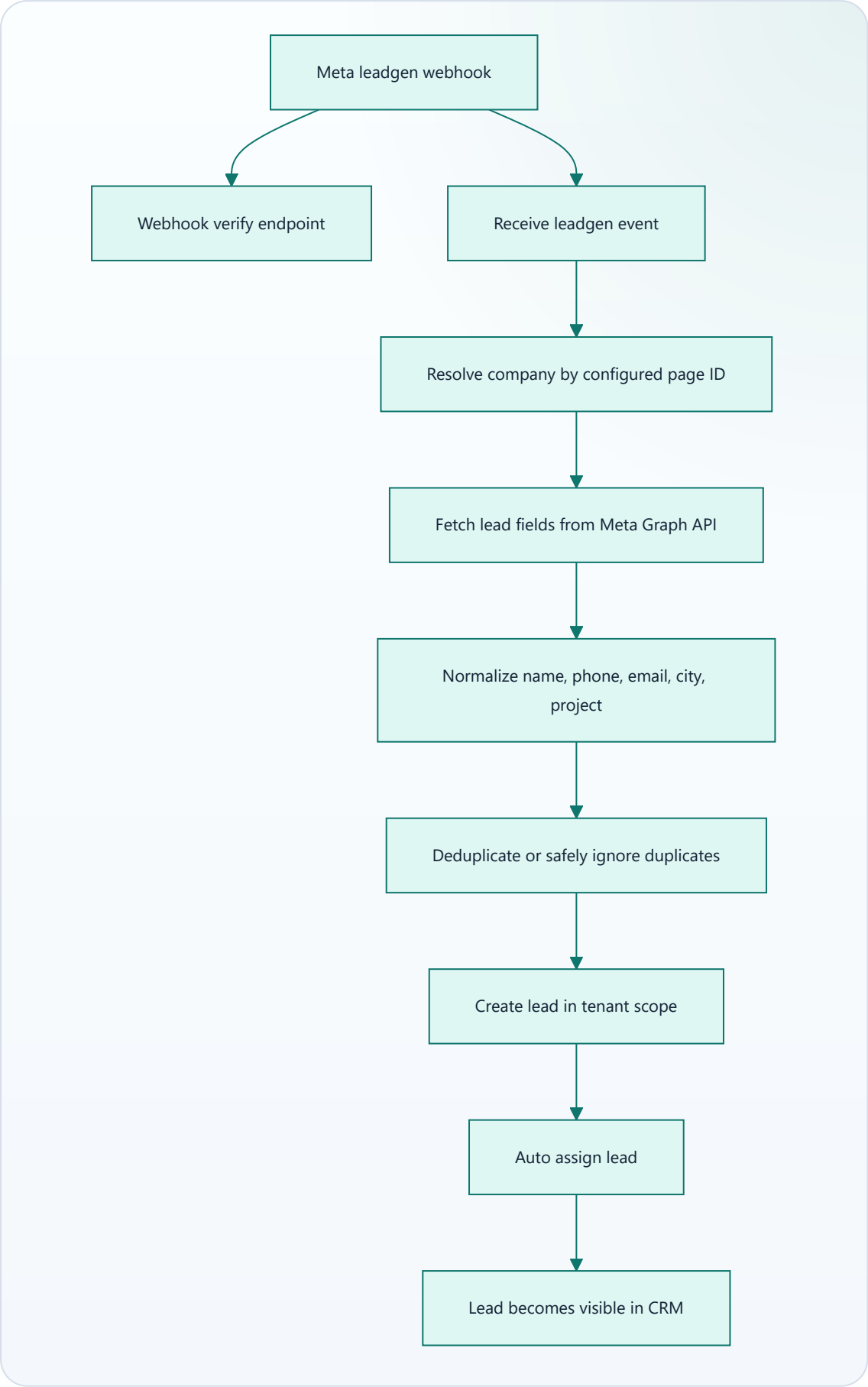
What runs through the realtime layer

- Room and direct messaging
- Typing, delivered, seen, and room-read receipts
- Incoming and outgoing call signaling
- Admin request alerts
- Inventory approval alerts
- Lead payment approval alerts

Primary source anchors

- `backend/src/server.js`
- `backend/src/socket/chat.socket.js`
- `backend/src/services/chatRoom.service.js`
- `backend/src/services/chatCall.service.js`
- `frontend/src/modules/chat/TeamChat.jsx`
- `frontend/src/context/chatNotificationProvider.jsx`
- `mobile/src/modules/chat/ChatConversationScreen.tsx`
- `mobile/src/modules/chat/CallScreen.tsx`
- `mobile/src/context/RealtimeAlertsContext.tsx`

9. Meta Webhook to Lead Pipeline



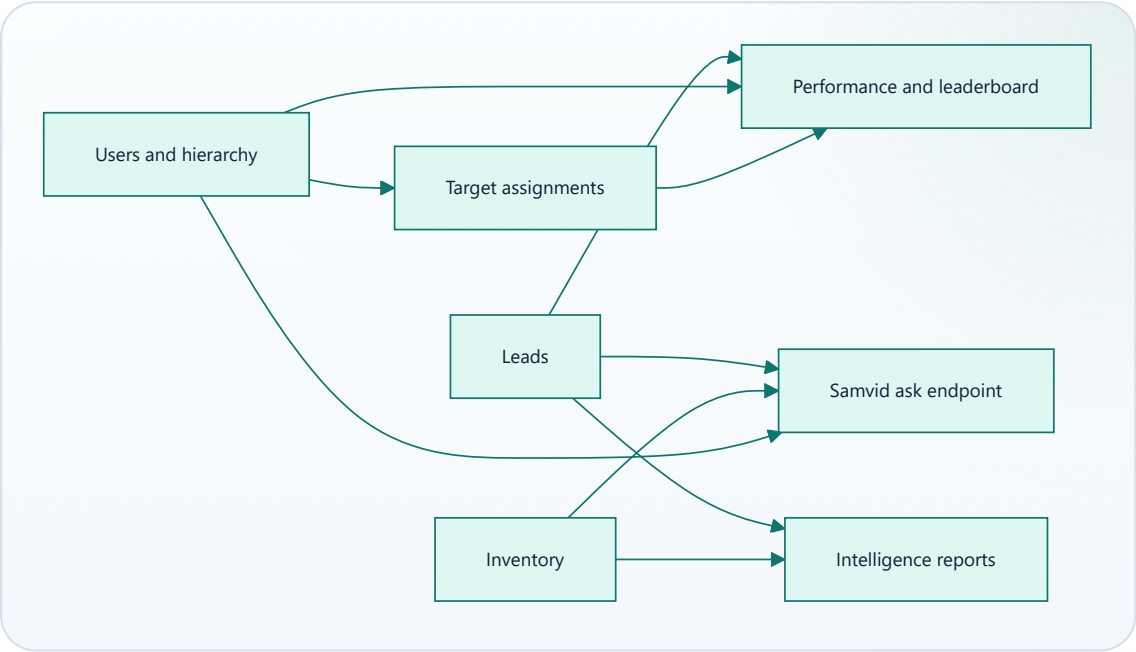
Why this matters

- A single webhook surface can route events to the correct tenant by page ownership.
- Company-scoped page configuration decides which tenant receives a Meta lead.
- New Meta leads immediately join the same lead workflow as manual and bulk-created leads.

Primary source anchors

- `backend/src/routes/webhook.routes.js`
- `backend/src/controllers/webhook.controller.js`
- `backend/src/services/leadAssignment.service.js`
- `backend/src/controllers/saas.controller.js`

10. Targets, Reports, and Samvid Intelligence



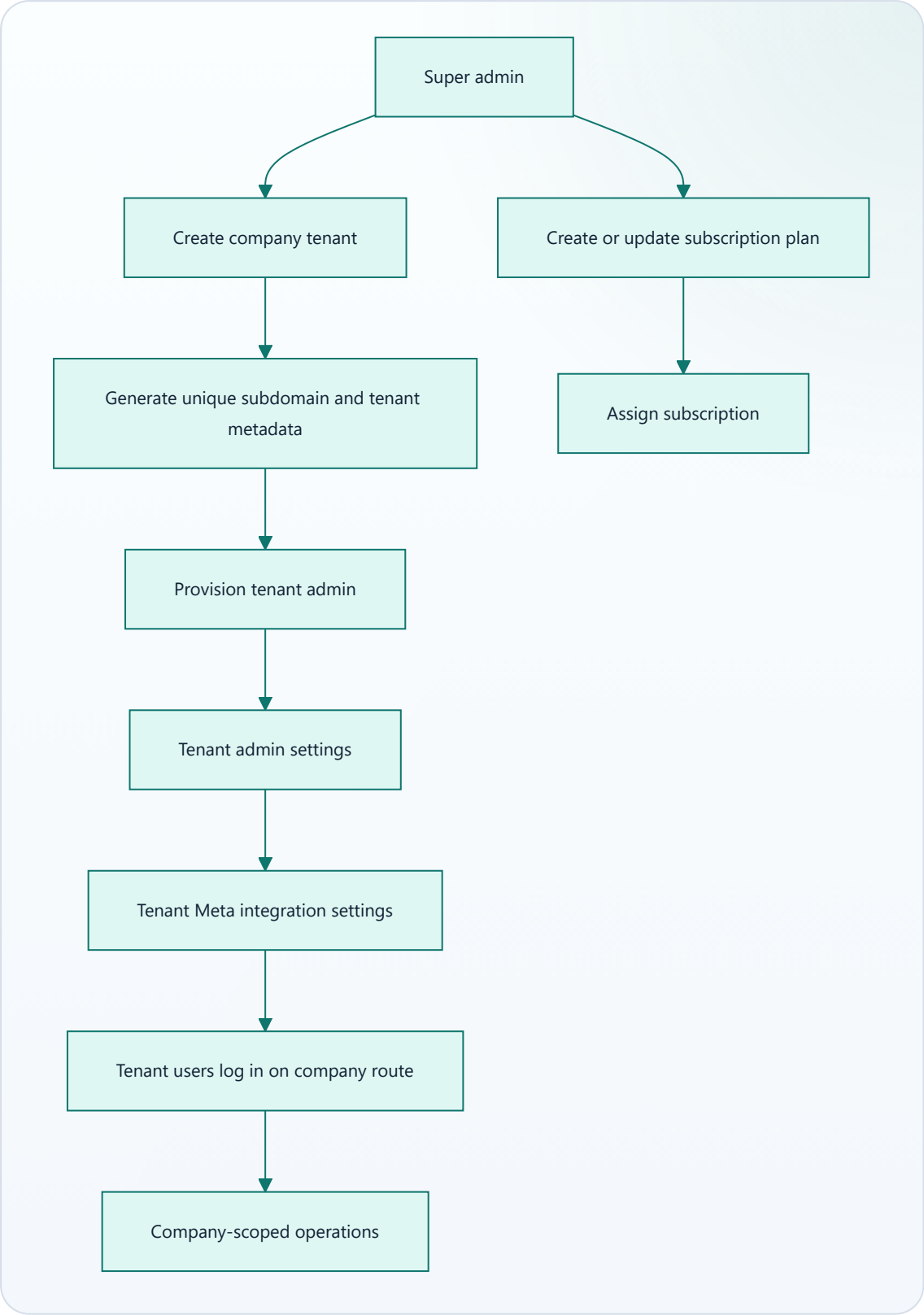
Functional view

- Targets are assigned through hierarchy-aware target APIs.
- Performance screens compare target achievement with live lead activity.
- Reports aggregate lead funnel, aging, inventory, and team output.
- Samvid queries the same live business data and returns intent-based snapshots.

Primary source anchors

- `backend/src/routes/target.routes.js`
- `backend/src/controllers/target.controller.js`
- `backend/src/routes/samvid.routes.js`
- `backend/src/controllers/samvid.controller.js`
- `frontend/src/modules/reports/Performance.jsx`
- `frontend/src/modules/reports/IntelligenceReports.jsx`
- `mobile/src/modules/reports/PerformanceScreen.tsx`
- `mobile/src/modules/chat/SamvidBotScreen.tsx`

11. SaaS and Tenant Administration Flow



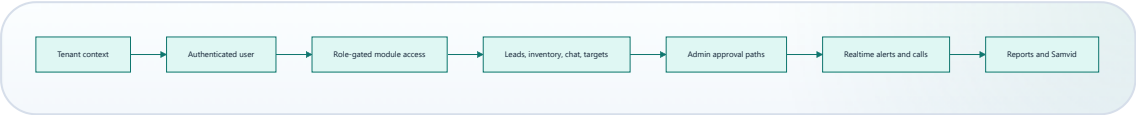
Operational view

- `SUPER_ADMIN` manages tenants, plans, analytics, and usage.
- Tenant admins manage their own settings and Meta integration.
- Once a tenant is configured, normal web and mobile users operate within that company scope.

Primary source anchors

- `backend/src/routes/saas.routes.js`
- `backend/src/controllers/saas.controller.js`
- `backend/SAAS_ARCHITECTURE.md`
- `frontend/src/modules/admin/SuperAdminPanel.jsx`
- `frontend/src/services/saasService.js`

12. End-to-End Summary



Samvid OS is essentially a tenant-aware operating system for sales teams:

- tenant resolution decides data ownership
- auth and role rules decide access
- lead and inventory workflows drive daily operations
- realtime events keep teams and admins synchronized
- reports, targets, and Samvid turn operations into insight